



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023-2024)

Test Pattern

NEET (UG)

MINOR

18-06-2023

PRE-MEDICAL : ENTHUSIAST ADVANCE COURSE PHASE : MEB-I (A)

IMPORTANT NOTE : Students having 8 digits **Form No.** must fill two zero before their **Form No.** in OMR. For example, if your **Form No.** is 12345678, then you have to fill **0012345678**.

Test Booklet Code

This Booklet contains **28** pages.

E5

Do not open this Test Booklet until you are asked to do so.

Important Instructions :

- The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on ORIGINAL Copy carefully with **blue/black** ball point pen only.
- The test is of **3 hours 20 minutes** duration and the Test Booklet contains **200** multiple-choice questions (four options with a single correct answer) from **Physics, Chemistry and Biology (Biology-I and Biology-II)**. **50** questions in each subject are divided into **two Sections (A and B)** as per details given below :
 - Section A** shall consist of **35 (Thirty-five)** Questions in each subject (Question Nos - 1 to 35, 51 to 85, 101 to 135 and 151 to 185). All questions are compulsory.
 - Section B** shall consist of **15 (Fifteen)** questions in each subject (Question Nos - 36 to 50, 86 to 100, 136 to 150 and 186 to 200). In Section B, a candidate needs to **attempt any 10 (Ten)** questions out of **15 (Fifteen)** in each subject. **Candidates are advised to read all 15 questions in each subject of Section B** before they start attempting the question paper. In the event of a candidate attempting more than ten questions, **the first ten questions answered by the candidate shall be evaluated.**
- Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one mark** will be deducted from the total scores. **The maximum marks are 720.**
- Use **Blue/Black Ball Point Pen only** for writing particulars on this page/markings responses on Answer Sheet.
- Rough work is to be done in the space provided for this purpose in the Test Booklet only.
- On completion of the test, the candidate **must hand over the Answer Sheet (ORIGINAL and OFFICE Copy) to the Invigilator** before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
- The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Form No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
- Use of white fluid for correction is **NOT** permissible on the Answer Sheet.
- Each candidate must show on-demand his/her Allen ID Card to the Invigilator.
- No candidate, without special permission of the Invigilator, would leave his/her seat.
- The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign (with time) the Attendance Sheet **twice**. **Cases, where a candidate has not signed the Attendance Sheet second time, will be deemed not to have handed over the Answer Sheet and dealt with as an Unfair Means case.**
- Use of Electronic/Manual Calculator is prohibited.
- The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per the Rules and Regulations of this examination.
- No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.**
- The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.
- Compensatory time of one hour five minutes will be provided for the examination of three hours and 20 minutes duration, whether such candidate (having a physical limitation to write) uses the facility of scribe or not.

Name of the Candidate (in Capitals) : _____

Form Number : in figures _____

: in words _____

Centre of Examination (in Capitals) : _____

Candidate's Signature : _____ Invigilator's Signature : _____

TC: @Chalnaayaaar

Your Target is to secure Good Rank in Pre-Medical 2024

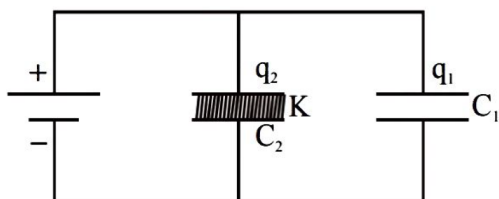
Topic : ELECTROSTATICS, CURRENT ELECTRICITY, CAPACITOR, GRAVITATION (COMPLETE), SPRING SYSTEM (SHM) TO WAVE MOTION, [BACK UNIT] : ELECTROSTATICS AND CURRENT ELECTRICITY

SECTION-A (PHYSICS)

1. A charged parallel plate capacitor of distance (d) has U_0 energy. A slab of dielectric constant (K) and thickness (d) is then introduced between the plates of the capacitor. The new energy of the system is given by :

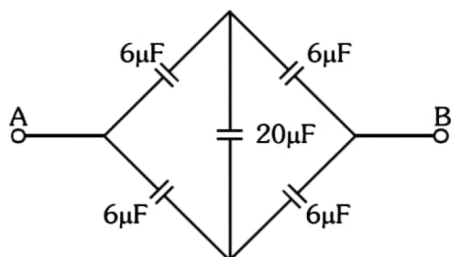
- (1) KU_0
 (2) K^2U_0
 (3) $\frac{U_0}{K}$
 (4) $\frac{U_0}{K^2}$

2. In the given figure, two identical capacitor C_1 and C_2 are connected to battery. Air is filled between the plates of C_1 and a dielectric is filled between the plates of C_2 then :



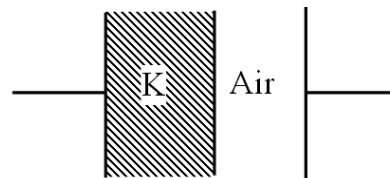
- (1) $q_1 < q_2$ (2) $q_1 > q_2$
 (3) $q_1 = q_2$ (4) None of these

3. The effective capacity of the network between terminals A and B is :



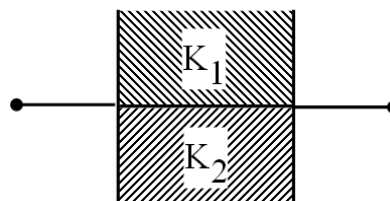
- (1) $6 \mu F$
 (2) $20 \mu F$
 (3) $3 \mu F$
 (4) $10 \mu F$

4. A parallel plate capacitor has capacitance C. Half of space between the plates is filled with dielectric of dielectric constant K as shown in figure. The new capacitance is :-



- (1) $C \left(\frac{K}{K+1} \right)$ (2) $C \left(\frac{2K}{K+1} \right)$
 (3) $\frac{2C}{K+1}$ (4) $C \left(1 + \frac{K}{2} \right)$

- 5.



A capacitor of plate area A and distance d is filled with two dielectric of dielectric constant $K_1 = 6$ and $K_2 = 4$. New capacitance will be :-

- (1) $4 \frac{\epsilon_0 A}{d}$ (2) $4.8 \frac{\epsilon_0 A}{d}$
 (3) $5 \frac{\epsilon_0 A}{d}$ (4) $2.4 \frac{\epsilon_0 A}{d}$

6. Then $30 \mu C$ charge is given to an isolated metal sphere of capacitance $5 \mu F$. If this sphere is now connected to another isolated metal sphere by a conducting wire (at very large distance) of total charge $50 \mu C$ and capacity $10 \mu F$ then common potential and the ratio of final charges on spheres is :

- (1) $\frac{16}{3}$ volt, $\frac{3}{5}$
 (2) $\frac{3}{16}$ volt, $\frac{1}{2}$
 (3) $\frac{16}{3}$ volt, $\frac{1}{2}$
 (4) $\frac{16}{5}$ volt, $\frac{2}{5}$

7. Energy stored between the plates of parallel plate capacitor of area 'A' and separated by a distance d. If electric field between its plates is E :

(1) $\frac{1}{2} \epsilon_0 E^2 \frac{A}{d}$ (2) $\frac{1}{2} \epsilon_0 \frac{d}{E^2 A}$
 (3) $\frac{1}{2} \epsilon_0 E^2 A d$ (4) $\epsilon_0 E A d$

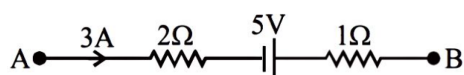
8. The electrostatic force between the metal plates of an parallel plate capacitor 'C' having a area A and distance between plates d. (Capacitor is connected with a battery of emf V) is :

- (1) Independent of the distance between the plates
 (2) Linearly proportional to the distance between the plates
 (3) Proportional to the square root of the distance between the plates
 (4) Inversely proportional to the square of the distance between the plates

9. Which of the following statements is false ?

- (1) Kirchoff's second law represents energy conservation.
 (2) Ohm's law is not a fundamental law of nature.
 (3) If a wire is stretched to 'n' times of it's original length, its new resistance will be n^2 times.
 (4) Temperature coefficient (α) of semiconductor is positive.

10. The potential difference ($V_B - V_A$) between the points A & B in the given figure is:-



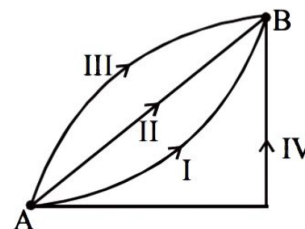
- (1) +6 V
 (2) +9 V
 (3) +4 Volt
 (4) -4 Volt

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11. An electric dipole consists of two opposite charges each of magnitude $1.0 \mu\text{C}$, separated by a distance of 2.0 cm. The dipole is placed in an external electric field of $1.0 \times 10^5 \text{ N/C}$. The maximum torque on the dipole is:

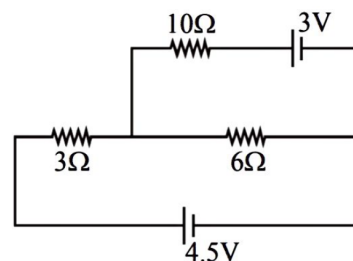
- (1) $0.2 \times 10^{-3} \text{ N-m}$
 (2) $1.0 \times 10^{-3} \text{ N-m}$
 (3) $2.0 \times 10^{-3} \text{ N-m}$
 (4) $4.0 \times 10^{-3} \text{ N-m}$

12. Under the influence of charge, a point charge 'q' is carried along different paths from a point 'A' to point 'B' then work done will be:-



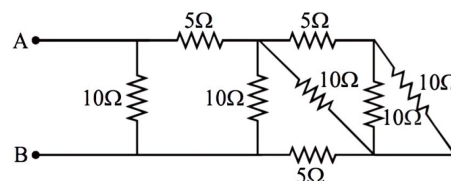
- (1) Maximum for path four.
 (2) Maximum for path one
 (3) Equal for all paths
 (4) Minimum for path three.

13. The current through the 10Ω resistor is



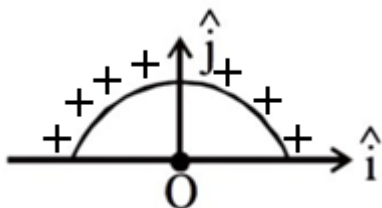
- (1) zero (2) 1 A (3) 2 A (4) 3 A

14. Equivalent resistance between terminals A and B



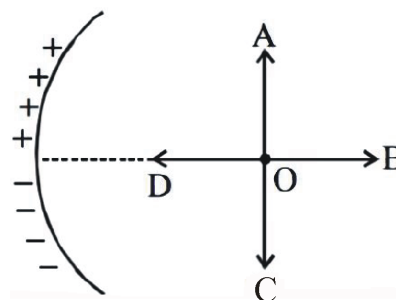
- (1) 2.5Ω (2) 15Ω
 (3) 5Ω (4) 10Ω

15. A thin semi-circular ring of radius 'r' is uniformly charged with linear charge density λ . The net field $\vec{E}$ at centre 'O' is :-

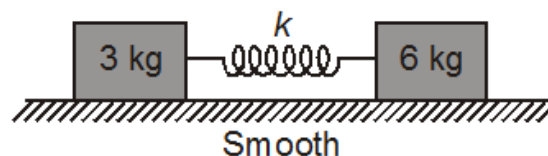


- (1) $\frac{K\lambda}{r}(-\hat{j})$
 - (2) $\frac{\sqrt{2}K\lambda}{r}(-\hat{j})$
 - (3) $\frac{\lambda}{2\pi\epsilon_0 r}(-\hat{j})$
 - (4) $\frac{q}{2\pi^2\epsilon_0 r^2}(-\hat{j})$
16. Select the correct statement(s) :-
- (a) The electric lines of force are always closed curves.
 - (b) Electric potential depends on reference point.
 - (c) Electric lines of force are perpendicular to equipotential surface.
 - (d) Electric field is conservative in nature so work done between two points is path independent depends only on the initial and final position.
- (1) Only b
 - (2) a, b & c
 - (3) b, c & d only
 - (4) a, b, c, d
17. Two point charges $-4Q$ and $9Q$ are placed at a distance 2m from each other. The position at which net electric field is zero from the charge $-4Q$ is:-
- (1) 3 m
 - (2) 4 m
 - (3) 2 m
 - (4) 1 m

18. The linear charge density on upper half of a segment of ring is ' λ ' and at lower half, it is $-\lambda$. The direction of electric field at centre 'O' of ring is:-



- (1) along OA
 - (2) along OB
 - (3) along OC
 - (4) along OD
19. Which one of the following statement is true ?
- (1) Both light and sound waves in air are transverse.
 - (2) The sound waves in air are longitudinal while the light waves are transverse.
 - (3) Both light and sound waves in air are longitudinal
 - (4) Both light and sound waves can travel in vacuum
20. Two blocks of masses 3 kg and 6 kg oscillate with a light spring $k = 200$ N/m as shown. The time period is

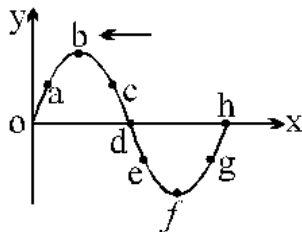


- (1) $\frac{\pi}{20}$ s
- (2) $\frac{\pi}{10}$ s
- (3) $\frac{\pi}{5}$ s
- (4) 2π s

21. The period of a simple pendulum is doubled, when :-

- (1) Its length is doubled
- (2) The mass of the bob is doubled
- (3) Its length is made four times
- (4) The mass of the bob and the length of the pendulum are doubled

22. The figure represents the instantaneous picture of a transverse harmonic wave traveling along the negative x-axis. Choose the correct alternative(s) related to the movement of the nine points shown in the figure.



The points moving downwards is

- (1) g (2) a (3) e (4) h

23. The equation for progressive wave is $y = 0.005 \cos \left(62.8t + 3.14x + \frac{\lambda}{3} \right)$ its periodic time T and wavelength λ are :-

- (1) 3.14s, 1m
- (2) 1s, 1m
- (3) 0.1s, 2m
- (4) 0.1s, 1m

24. Newton's formula for the velocity of sound in gases is :

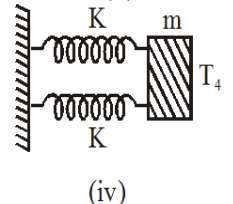
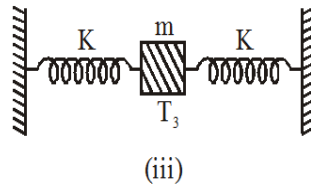
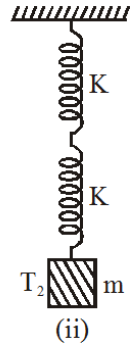
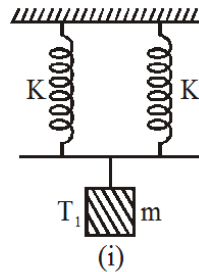
- (1) $v = \sqrt{\frac{2p}{\rho}}$
- (2) $v = \sqrt{\frac{p}{\rho}}$
- (3) $v = \sqrt{\frac{p}{\rho}}$
- (4) $v = \frac{3}{2} \sqrt{\frac{p}{\rho}}$

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PHASE - MEB

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25. A mass m is attached to two spring of same force constants, in accordance with the following four configurations. Answer the questions given below. Let T_1 , T_2 , T_3 and T_4 respectively be the time periods in the configurations (i), (ii), (iii) and (iv) :-



In which case the time period is largest.

- (1) (i) (2) (ii) (3) (iii) (4) (iv)

26. Simple pendulum of length ℓ and having a bob of mass M is suspended in a car. The car is moving on a circular track of radius R with uniform speed V . If the pendulum makes small oscillations then what is its time period :-

- (1) $2\pi\sqrt{\ell/g}$
- (2) $2\pi\sqrt{\frac{\ell}{\sqrt{g^2 + \frac{V^4}{R^2}}}}$
- (3) $2\pi\sqrt{\frac{\ell}{g^2 + \frac{V^4}{R^2}}}$
- (4) $2\pi\sqrt{\frac{R}{g}}$

27. A steel wire 0.72 m long has a mass of 5.0×10^{-3} kg. If the wire is under tension of 60 N, what is the speed of the transverse waves on the wire ?

- (1) 186 m/s (2) $\sqrt{2} \times 93$ m/s
- (3) 93 m/s (4) 45 m/s

28. The type of waves that can be propagated through solid is :-

- (1) Transverse (2) Longitudinal
- (3) Both (1) and (2) (4) None of these

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29. Following is given the equation of a traveling wave (all in SI units)

$$y = (0.02)\sin 2\pi(10t - 5x)$$

Match the followings:

List I

List II

(only magnitude in SI units)

- (A) Speed of wave (i) 10
(B) Frequency of wave (ii) 0.4π
(C) Wavelength of wave (iii) 2
(D) Maximum particle speed (iv) 0.2

- (1) A $\rightarrow$ (iv) B $\rightarrow$ (ii) C $\rightarrow$ (i) D $\rightarrow$ (iii)
(2) A $\rightarrow$ (iii) B $\rightarrow$ (ii) C $\rightarrow$ (i) D $\rightarrow$ (iv)
(3) A $\rightarrow$ (iv) B $\rightarrow$ (ii) C $\rightarrow$ (iii) D $\rightarrow$ (i)
(4) A $\rightarrow$ (iii) B $\rightarrow$ (i) C $\rightarrow$ (iv) D $\rightarrow$ (ii)

30. A body weighs 144 N at the surface of earth. When it is taken to a height of $h = 3R$, where R is radius of earth, its weight would be :

- (1) 48 N (2) 36 N
(3) 16 N (4) 9 N

31. How much deep inside the earth (radius R) should a man go, so that his weight becomes one fourth of that on the earth's surface ?

- (1) $\frac{R}{4}$ (2) $\frac{R}{2}$
(3) $\frac{3R}{4}$ (4) None of these

32. A planet is revolving round the sun. Its distance from the sun at Apogee is r_A and that at Perigee is r_P . The mass of planet and sun is m and M respectively. v_A and v_P is the velocity of planet at Apogee and Perigee respectively and T is the time period of revolution of planet round the sun.

(a) $T^2 = \frac{\pi^2}{2Gm}(r_A + r_P)^2$

(b) $T^2 = \frac{\pi^2}{2GM}(r_A + r_P)^3$

(c) $v_A r_A = v_P r_P$

(d) $v_A < v_P$, $r_A > r_P$

- (1) a, b, c (2) a, b, d (3) b, c, d (4) All

33. A geostationary satellite is orbiting the Earth at a height of $6R$ above the surface of the earth, R being the radius of the Earth. The time period of another satellite, at a height of $2.5R$ from the surface of the Earth, is

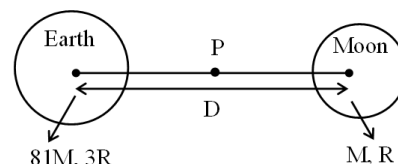
- (1) $6\sqrt{2}$ hour (2) 6 hour
(3) $5\sqrt{2}$ hour (4) 10 hour

34. Find the work required in taking a mass m from earth's surface to a height equal to twice the earth's radius.

(M = mass of earth, R = radius of earth)

- (1) $\frac{GMm}{2R^2}$ (2) $\frac{GMm}{3R^2}$
(3) $\frac{GMm}{4R^2}$ (4) $\frac{GMm}{R^2}$

35. Find the gravitational potential at the mid point (P) of the line joining earth and moon as shown in above figure.



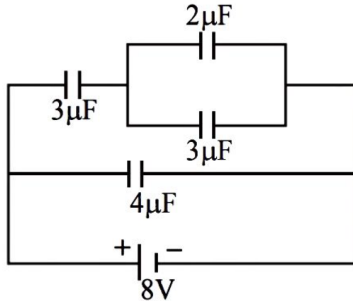
- (1) $\frac{-82GM}{D}$
(2) $\frac{-28GM}{D}$
(3) $\frac{-164GM}{D}$
(4) $\frac{-GM}{27D}$

SECTION-B (PHYSICS)

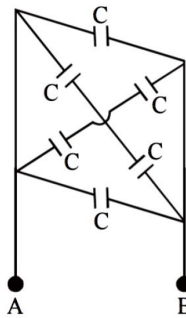
36. The energy density in a parallel plate capacitor is given as $2.1 \times 10^{-9} \text{ J/m}^3$. The value of the electric field in the region between the plates is:

- (1) 2.1 NC^{-1}
(2) 21.6 NC^{-1}
(3) 72 NC^{-1}
(4) 8.4 NC^{-1}

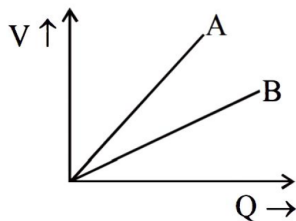
37. The charge on the capacitor of capacitance $2\mu\text{F}$ is:



- (1) $4.5\mu\text{C}$ (2) $6.0\mu\text{C}$
 (3) $7\mu\text{C}$ (4) $30\mu\text{C}$
38. The resultant capacity between the points A and B in the adjoining circuit will be :



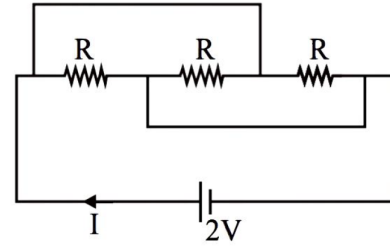
- (1) C (2) $2C$ (3) $3C$ (4) $4C$
39. The graph shows the variation of voltage V across the plates of two capacitors A and B with charge ' Q '. Relation between capacitance of A and B is:



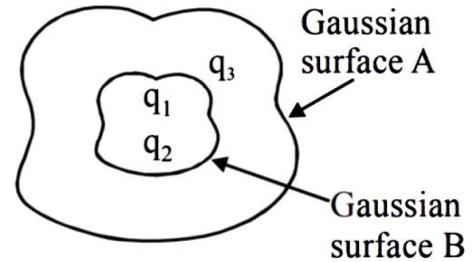
- (1) $C_A > C_B$ (2) $C_A = C_B$
 (3) $C_A < C_B$ (4) None of these
40. A rectangular carbon block has dimensions $1.0\text{ cm} \times 1.0\text{ cm} \times 50\text{ cm}$. The resistance measured between the two square ends is. (Resistivity of carbon is $3.5 \times 10^{-5}\Omega\text{-m}$)

- (1) 17.5Ω (2) 0.175Ω
 (3) 1.75Ω (4) $\frac{1000}{175}\Omega$

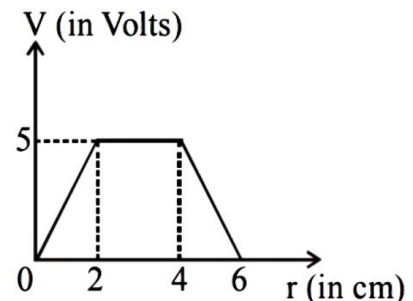
41. Three equal resistance each of $R = 1\Omega$ are connected as shown in figure. Internal resistance of cell is zero. The value of current I will be



- (1) 2A (2) 3A
 (3) 6A (4) None of these
42. The electric flux for gaussian surface 'A' that enclose the charged particles in free space is (given $q_1 = -14\text{ nC}$, $q_2 = 78.85\text{ nC}$, $q_3 = -56\text{ nC}$) :-

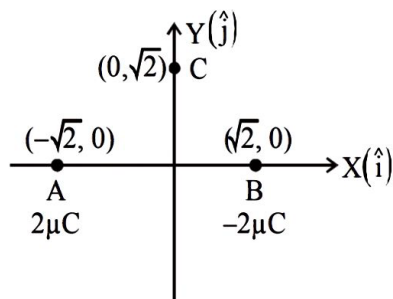


- (1) $10^3\text{ N m}^2\text{ C}^{-1}$
 (2) $6.32 \times 10^3\text{ N m}^2\text{ C}^{-1}$
 (3) $10^3\text{ C N}^{-1}\text{ m}^{-2}$
 (4) $6.32 \times 10^3\text{ C N m}^2$
43. The variation of potential with distance ' r ' from a fixed point is shown in figure. The electric field at $r = 5\text{ cm}$ is :-



- (1) 2.5 V/cm (2) -2.5 V/cm
 (3) $-\frac{2}{5}\text{ V/cm}$ (4) $\frac{2}{5}\text{ V/cm}$

44. Two point charges $2\mu\text{C}$ and $-2\mu\text{C}$ are placed at points A and B as shown in figure electric field intensity at point C is (All the distances are measured in meter) :-



- (1) $\frac{900}{\sqrt{2}}(\hat{i})\frac{\text{N}}{\text{C}}$
 (2) $\frac{9000}{2}(\hat{i})\frac{\text{N}}{\text{C}}$
 (3) $\frac{9000}{\sqrt{2}}(-\hat{i})\frac{\text{N}}{\text{C}}$
 (4) $\frac{9000}{\sqrt{2}}(\hat{i})\frac{\text{N}}{\text{C}}$
45. In damped oscillations, the amplitude after 50 oscillations is $0.8 a_0$, where a_0 is the initial amplitude, then the amplitude after 150 oscillations is:
- (1) $0.512 a_0$
 (2) $0.280 a_0$
 (3) Zero
 (4) a_0
46. The equation of a wave travelling on a string is
 $y = 4\sin \frac{\pi}{2} \left(8t - \frac{x}{8} \right)$
 if x and y are in cm, then velocity of the wave is
- (1) 64 cm s^{-1} in $-x$ direction
 (2) 32 cm s^{-1} in $-x$ direction
 (3) 32 cm s^{-1} in $+x$ direction
 (4) 64 cm s^{-1} in $+x$ direction

47. The ratio of the velocity of sound in hydrogen ($\gamma = 7/5$) to that in helium ($\gamma = 5/3$) at the same temperature is :

- (1) $\sqrt{\frac{5}{42}}$
 (2) $\sqrt{\frac{5}{21}}$
 (3) $\frac{\sqrt{42}}{5}$
 (4) $\frac{\sqrt{21}}{5}$

48. A simple pendulum has time period T_1 . The point of suspension is now moved upward according to the relation $y = Kt^2$, ($K = 1 \text{ m/s}^2$) where y is the vertical displacement. The time period now becomes T_2 . The ratio of $\frac{T_1^2}{T_2^2}$ is :
 ($g = 10 \text{ m/s}^2$)

- (1) $\frac{6}{5}$ (2) $\frac{5}{6}$
 (3) 1 (4) $\frac{4}{5}$

49. Planet A and planet B have masses $4M$ and M and radius R and $2R$ respectively. Find the ratio of escape speed of A and B.

- (1) 1 : 1
 (2) 2 : 1
 (3) 1 : 2
 (4) $2\sqrt{2} : 1$

50. Two satellites A and B of same mass are moving around a planet at an altitude of $3R$ and $5R$ respectively. Select the correct statement.

- (1) Ratio of total energy of A and B is 5 : 3
 (2) Ratio of potential energy of A and B is 3 : 5
 (3) Ratio of total energy of A and B is 3 : 2
 (4) Ratio of kinetic energy of A and B is 2 : 3

Topic : p-BLOCK (COMPLETE), d & f-BLOCK (TILL TAUGHT), CLASSIFICATION, NOMENCLATURE, STRUCTURAL AND GEOMETRICAL ISOMERISM, [BACK UNIT] : PERIODIC TABLE

SECTION-A (CHEMISTRY)

51. Match the column correctly ?

Column-I		Column-II	
(a)	CO	(P)	Neutral
(b)	Sb ₂ O ₃	(Q)	Acidic
(c)	Bi ₂ O ₃	(R)	Amphoteric
(d)	P ₄ O ₁₀	(S)	Basic

	a	b	c	d
1.	P	Q	S	R
2.	P	R	S	Q
3.	R	S	Q	P
4.	S	R	P	Q

52. Which of the following order of EA is correct ?

- (1) Ne > Be > N (2) Ne < Be > N
(3) Ne < Be < N (4) Ne > Be = N

53. In which of following distance between nuclei is maximum :-

- (1) CsF (2) KI (3) CsI (4) LiI

54. Size of Mo is very similar to W due to -

- (1) Difference of atomic number by one
(2) Contraction in size in first transition series element.
(3) Lanthanide contraction
(4) actinide contraction

55. The ionization potentials of Li and K are 5.4 and 4.3 eV respectively. The ionization potential of Na will be :-

- (1) 9.7 eV (2) 1.1 eV
(3) 4.9 eV (4) cannot be calculated

56. An element whose IUPAC name is ununtrium (UUT) belongs to :

- (1) s - block (2) p - block
(3) d - block (4) f - block

57. The nitrogen oxide that dose not contain N–N bond are—

- (1) N₂O (2) N₂O₃
(3) N₂O₄ (4) N₂O₅

58. In the ostwald's process nitric acid is prepared by the catalytic oxidation of :

- (1) N₂ (2) NH₃
(3) N₂O₅ (4) NO₂

59. $P + Cl_2 \rightarrow A$, $P + \text{Excess } Cl_2 \rightarrow B$
Hydrolysis product of A and B respectively.

- (1) H₃PO₂, H₃PO₃
(2) H₃PO₃, H₃PO₄
(3) H₃PO₄, H₃PO₃
(4) H₃PO₂, H₃PO₄

60. XeF₂ reacts with PF₅ to give:

- (1) XeF₆ (2) [XeF]⁺[PF₆]⁻
(3) XeF₄ (4) [PF₄]⁺[XeF₃]⁻

61. **Assertion (A)** : Reducing property of dioxide decreases as SO₂ > SeO₂ > TeO₂.

Reason (R) : SO₂ is oxidising while TeO₂ is reducing agent.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

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PHASE - MEB

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62. Which is incorrectly given according to order indicated?

- (1) $F_2 > Cl_2 > Br_2 > I_2$; Oxidising power
- (2) $HI > HBr > HCl > HF$: Acidic strength
- (3) $F_2 > Cl_2 > Br_2 > I_2$: Bond dissociation enthalpy
- (4) $HF > HI > HBr > HCl$: Boiling point

63. Which of the following statements are correct for allotropes.

- (A) Ozone is allotrope of oxygen.
- (B) Fullerene is purest form of allotrope of carbon.
- (C) The most stable allotrope of sulphur at room temperature is rhombic sulphur.
- (D) The order of catenation is $C \gg Si > Ge \approx Sn$, so carbon forms maximum number of allotropes.
- (E) Nitrogen do not form allotropes but halogens forms several allotropes.

Choose the correct answer from the options given below :

- (1) B, C, D only
- (2) A, B, C, D only
- (3) A, B, D only
- (4) A, B, C only

64. Match list-I with list-II.

List-I		List-II	
(A)	B_4C	(I)	Interstitial carbide
(B)	Al_4C_3	(II)	Gives propyne on hydrolysis
(C)	Mg_2C_3	(III)	Gives propyne on hydrolysis
(D)	Fe_3C	(IV)	Used as abrasive

Choose the correct answer from options given below :

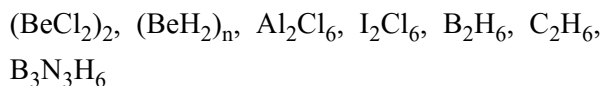
- (1) A-II, B-III, C-I, D-IV
- (2) A-IV, B-II, C-III, D-I
- (3) A-II, B-III, C-IV, D-I
- (4) A-II, B-I, C-III, D-IV

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65. Nitrolium is

- (1) $CaC_2 + N_2$
- (2) $Ca(CN)_2 + C$
- (3) $CaCN_2 + C$
- (4) $Ca + (CN)_2$

66. Amongst the following total number of planar species are :



- (1) 3
- (2) 4
- (3) 7
- (4) 2

67. **Assertion** : Bleaching action O_3 is permanent while that of SO_2 is temporary.

Reason : O_3 is a dry bleaching agent.

- (1) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (2) Assertion is True but the Reason is False.
- (3) Assertion is False but Reason is True.
- (4) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

68. Incorrect order of size is :

- (1) $Cu > Zn$
- (2) $Sc > Ti$
- (3) $Co \approx Ni$
- (4) $Zr \approx Hf$

69. If ΔH_{eg} of $A(g) = -y$ kJ/mol and ΔH_{IE} of $A(g) = x$ kJ/mol, then change in enthalpy (ΔH) for the process $A_{(g)}^- \rightarrow A_{(g)}^+$ is :

- (1) $x - y$
- (2) $x + y$
- (3) $-x + y$
- (4) $-x - y$

70. In which of the following compound central atom has largest ionic radii :

- (1) Cr_2O_3
- (2) CrO
- (3) $K_2Cr_2O_7$
- (4) CrO_5

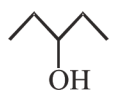
71. If electronic configuration of $X_{(g)}$ is $1s^2 2s^2$ and $X_{(g)}^+$ is $1s^2 2s^1$ then compare ionisation energy of $X(g)$ and $X_{(g)}^+$:

- (1) $X_{(g)}^+ > X_{(g)}$
- (2) $X_{(g)} > X_{(g)}^+$
- (3) $X_{(g)}^+ \approx X_{(g)}$
- (4) Can't be compared

72. Amongst N, O, Ne, Na and Na^+ , species having highest and lowest ionisation energy respectively.

- (1) O, Na^+ (2) Na^+ , Na
(3) Na, Ne (4) N, Ne

73. Find correct relationship :-

- (1)  and  Position isomers
(2)  and  Geometrical isomers
(3)  and  Chain isomers
(4)  and  Ring chain isomers

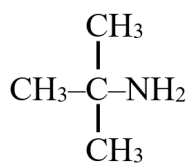
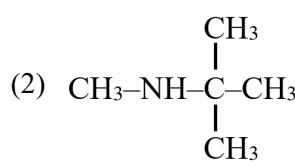
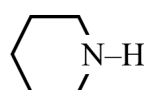
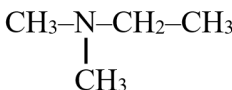
74. Which of the following is an example of functional group isomer :-

- (1) $\text{CH}_3\text{C}(=\text{O})\text{CH}_3$, CH_3CHO
(2) $\text{CH}_3\text{CH}_2\text{OH}$, CH_3CHO
(3) $\text{CH}_3\text{CH}_2\text{CHO}$, $\text{CH}_3\text{C}(=\text{O})\text{CH}_3$
(4) $\text{CH}_3\text{C}(=\text{O})\text{CH}_3$, $\text{CH}_3\text{CH}_2\text{OH}$

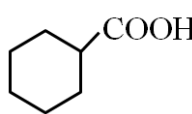
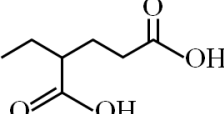
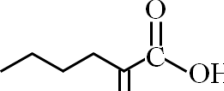
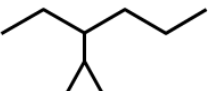
75. $\text{CH}_3\text{NHCH}_2\text{CH}_3$ and $(\text{CH}_3)_3\text{N}$ show which type of isomerism :-

- (1) Position
(2) Functional
(3) Chain
(4) None

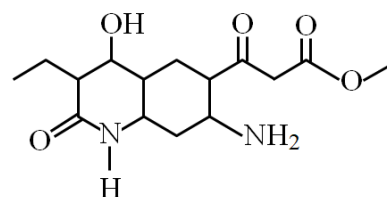
76. Which is tertiary amine ?

- (1)  (2) 
(3)  (4) 

77. Shortest parent carbon chain is in which of the following :-

- (1)  (2) 
(3)  (4) 

78. Which functional group is absent in the following compound :-

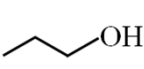
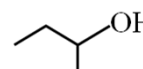
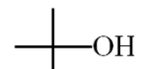
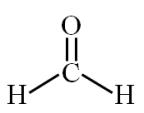
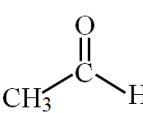
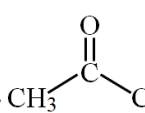


- (1) Ketone (2) 1° amine
(3) 2° alcohol (4) 2° amine

79. Correct priority for selecting principal functional group according to IUPAC?

- (1) $-\text{SO}_3\text{H} > -\text{COOH} > -\text{COOR} > -(\text{CO})_2\text{O}$
(2) $-\text{CHO} > -\text{C}(=\text{O})- > -\text{CN} > -\text{NH}_2$
(3) $-\text{COOR} > -\text{COX} > -\text{CN} > -\text{OH}$
(4) $-\text{NH}_2 > -\text{OH} > -\text{SH} > -\text{CN}$

80. Homologous compounds are :

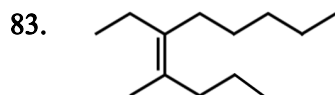
- (1) , , 
(2) , , 
(3) CH_3NH_2 , CH_3NHCH_3 , $(\text{CH}_3)_3\text{N}$
(4) CH_3COOH , $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$, $\text{CH}_3(\text{CH}_2)_4\text{COOH}$

81. Compound with maximum number of π bonds is :-

- (1) Tetra cyanoethylene (2) Benzene
(3) Naphthalene (4) Pyridine

82. Correct IUPAC name matched with compounds with following common names :-

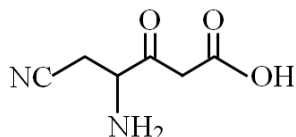
- (1) t-butyl alcohol $\equiv$ 1,1-Dimethyl ethanol
- (2) Allyl chloride $\equiv$ 1-Chloro-2-propene
- (3) Succinic acid $\equiv$ Butanedioic acid
- (4) Isobutyl alcohol $\equiv$ 2-methyl 2-propanol



How many carbon atoms are present in the parent carbon chain of the above compound, according to IUPAC naming :-

- (1) 10 (2) 7 (3) 8 (4) 9

84. Correct IUPAC name is :-



- (1) 3-Amino-5-Carboxy-4-oxopentanenitrile
- (2) 4-Amino-5-Cyano-3-oxo pentanoic acid
- (3) 4-Amino-6-Cyano-3-oxo hexanoic acid
- (4) 4-Amino-3-oxo-5-nitrile pentanuic acid

85. How many primary amines ($R-NH_2$) are possible from $C_5H_{13}N$

- (1) 4 (2) 5 (3) 6 (4) 8

SECTION-B (CHEMISTRY)

86. **Assertion (A)** : Acidic nature of phosphinic acid is greater than phosphonic acid.

Reason (R) : Acidic nature increases with increase in electronegative of 'P' atom.

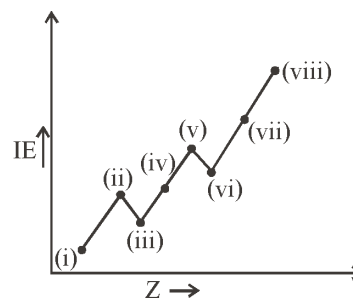
- (1) (A) and (R) both correct and R is the correct explanation of A.
- (2) (A) and (R) both correct but R is not correct explanation of A.
- (3) A is correct but R is incorrect.
- (4) A is incorrect but R is correct.

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PHASE - MEB

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87. A graph is plotted between IE of 2nd period elements and their atomic numbers as below :-



then which of the following option is **incorrect** ?

- (1) at (i) position Li is present
- (2) at (viii) position Ne is present
- (3) at (vi) position N is present
- (4) at (iii) position B is present

88. Which of the following statement is not correct :-

- (1) Hypophosphoric acid consists of a P-P linkage
- (2) Phosphorescence shown by phosphorous is due to oxidation of phosphorous
- (3) $PbSO_4$ is formed when PbS react with O_3 or H_2O_2
- (4) Silicones repel water due to strong Si-O-Si bonds

89. **Assertion:** Boric acid (H_3BO_3) and fluoboric acid (HF_4) both are mono basic acid in water.

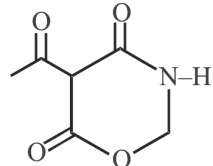
Reason: Both the acid are OH^- acceptor's rather than proton donor.

- (1) Both Assertion and Reason are correct and Reason is correct for the Assertion.
- (2) Both Assertion and Reason are correct but Reason is not correct for Assertion.
- (3) Assertion is correct but Reason is incorrect.
- (4) Assertion is incorrect but Reason is correct.

90. Which of the following halogen disproportionates in water ?

- (1) F_2 (2) Cl_2
- (3) I_2 (4) All three

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91. $\text{H}_3\text{PO}_3 \xrightarrow{-\text{O}} \text{X}$
 $\text{X} \xrightarrow{\Delta} \text{Y} + \text{PH}_3$
 Relation between basicity of X and Y is
 (1) Basicity of X > Y
 (2) Basicity of X = Y
 (3) Basicity of X < Y
 (4) Basicity is not defined for acids.
92. Select the correct statements from the following :
 (A) H_2SO_5 is peroxodisulphuric acid
 (B) $\text{H}_2\text{S}_2\text{O}_7$ (oleum) has one (S–O–S) linkage and two (O–H) bonds.
 (C) H_3PO_3 is phosphonic acid with two P–OH bonds.
 (D) $(\text{HPO}_3)_4$ is metaphosphoric acid, which exists in polymeric form only.
 (E) HOClO_3 is perchloric acid with one basicity.
 Choose the correct answer from the following options :
 (1) A, C only
 (2) A, B, D, E only
 (3) A, B, C, D, only
 (4) B, D, E only
93. **Assertion** : SiCl_4 undergoes hydrolysis but CCl_4 does not.
Reason : Silicon has vacant d-orbitals in its valence shell whereas carbon does not.
 (1) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
 (2) Assertion is True but the Reason is False.
 (3) Assertion is False but Reason is True.
 (4) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
94. Which of the following difference of ionisation energy will have largest value for Na.
 (1) $\text{IE}_2 - \text{IE}_1$ (2) $\text{IE}_3 - \text{IE}_2$
 (3) $\text{IE}_4 - \text{IE}_3$ (4) $\text{IE}_3 - \text{IE}_1$
95. Which of the following process is exothermic ?
 (1) $\text{P} \rightarrow \text{P}^+$
 (2) $\text{P} \rightarrow \text{P}^-$
 (3) $\text{P}^- \rightarrow \text{P}$
 (4) Both (1) and (2)
96. $\text{CH}_3-\text{CH} \begin{cases} \text{CHO} \\ \text{CHO} \end{cases}$ and $\begin{matrix} \text{CH}_2-\text{CH}_2 \\ | \quad | \\ \text{CHO} \quad \text{CHO} \end{matrix}$ are :-
 (1) Position isomer
 (2) Metamer
 (3) Chain isomer
 (4) Functional group isomer
97. The total number of isomeric geminal dihalides having the molecular formula $\text{C}_3\text{H}_6\text{Cl}_2$ is :-
 (1) 2 (2) 3 (3) 4 (4) 5
98. Identify which functional group is not present in the given compound :
- 
- (1) Ketone (2) Ester
 (3) Amide (4) Ether
99. How many cyclic structural isomers of C_5H_{10} ?
 (1) 4 (2) 5
 (3) 6 (4) 7
100. How many isomeric alkanes are possible from C_7H_{16} [Only structural isomers]
 (1) 7 (2) 8
 (3) 9 (4) 6

Topic : PRINCIPLE OF INHERITANCE (GENETICS) INTRODUCTION TO CYTOPLASMIC INHERITANCE, BIOMES, BIODIVERSITY, DEMOGRAPHY AND POLLUTION, COMPLETE ANIMAL HUSBANDRY (DOA), STRUCTURAL ORGANISATION IN ANIMALS (ANIMAL TISSUE), [BACK UNIT] : LIFE CYCLE OF ANGIOSPERM

SECTION-A (BIOLOGY-I)

101. Which of the following statement is correct?

- (1) Floral primordium is developed as a result of hormonal activities
- (2) Every flower has both male and female reproductive structures
- (3) Flower is a modified root
- (4) Floral organs are modified buds

102. If leaf tip cells has 28 chromosomes, what will be the number of chromosomes in middle layer, generative cell and male gametes ?

- (1) 14, 28, 14
- (2) 28, 14, 14
- (3) 14, 14, 28
- (4) 28, 28, 14

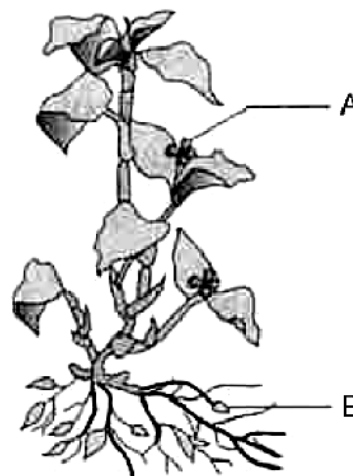
103. Chalaza representing the

- (1) Tip of the ovule
- (2) Base of the ovule
- (3) Both (1) & (2)
- (4) Stalk of the ovule

104. The wrong match regarding the morphology of following structures would be _____

- (1) Stamen – Microsporophyll
- (2) Carpel – Megasporophyll
- (3) Nucellus – Microsporangium
- (4) Ovule – Megasporangium

105. What are A and B in this figure?



- (1) A: Chasmogamous flower, B: Cleistogamous flower
- (2) A: Cleistogamous flower, B: Chasmogamous flower
- (3) A: Chasmogamous flower, B: Chasmogamous flower
- (4) A: Cleistogamous flower, B: Self-pollinated flower

106. Dioecious plants (papaya, date palm) etc. prevent

- (1) Autogamy but not geitonogamy
- (2) Geitonogamy but not autogamy
- (3) Both autogamy and geitonogamy
- (4) Neither autogamy nor geitonogamy

107. Primary endosperm nucleus (PEN) is formed by the fusion of

- (1) 2 polar nuclei + 1 synergid cell nucleus
- (2) 1 polar nucleus + 1 antipodal cell nucleus + 1 synergid cell nucleus
- (3) 2 polar nuclei + 1 male gamete nucleus
- (4) 2 antipodal cell nuclei + 1 male gamete nucleus

108. Which one of the following statements regarding post-fertilization development in flowering plants is incorrect?

- (1) Ovules develop into an embryo sac
- (2) Ovary develops into fruit
- (3) Zygote develops into an embryo
- (4) Central cell develops into endosperm

109. A typical dicotyledonous embryo, consists of an embryonal axis and two cotyledons. The portion of embryonal axis above the level of cotyledons is the :

- (1) Hypocotyl (2) Epicotyl
- (3) Epiblast (4) Radicle

110. Endosperm persists in the mature seeds of

- (1) Pea (2) Groundnut
- (3) Gram (4) Maize

111. **Statement I :** Pollen grains are rich in nutrients.

Statement II : In some cereals such as rice and wheat, pollen grains lose viability within 30 minutes of their release.

- (1) Statement-I and statement-II both are true
- (2) Statement-I is true and statement-II is false
- (3) Statement-I is false and statement-II is true
- (4) Statement-I and statement-II both are false

112. Given below are two statements :-

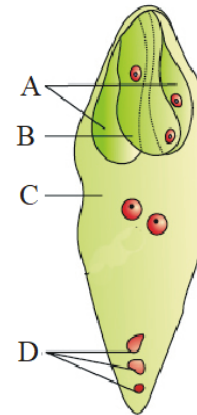
Statement-I :- Megaspore divides twice to form an eight nucleate embryo sac.

Statement-II :- Egg and synergids always lies near the micropylar end of ovule.

- (1) Statement-I and II both are correct.
- (2) Statement-I and II both are incorrect.
- (3) Only Statement-I is correct.
- (4) Only Statement-II is correct.

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113. Given below a diagram of mature embryo sac. Identify the parts labelled A, B, C and D and select correct option about them.



	Part-A	Part-B	Part-C	Part-D
(1)	Synergids	Egg	Central cell	Antipodals
(2)	Egg	Synergids	Central cell	Antipodals
(3)	Antipodals	Central cell	Synergids	Egg
(4)	Central cell	Egg	Synergids	Antipodals

114. Match List-I with List-II

List-I		List-II	
(a)	Tapetum	(i)	Provides nutrition to developing embryo
(b)	Nucellus	(ii)	Provides nutrition during seed germination
(c)	Endosperm	(iii)	Provides nutrition to embryo sac
(d)	Cotyledons	(iv)	Provides nutrition to developing microspore

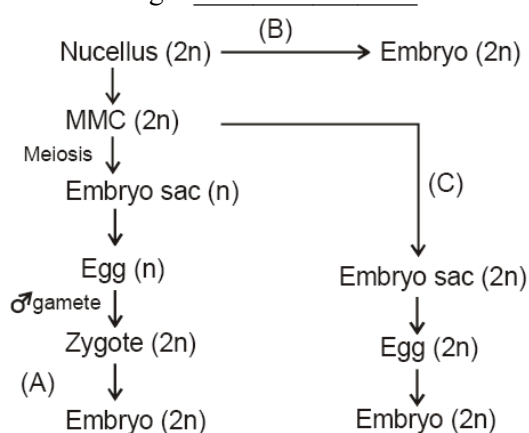
Choose the correct answer from the options given below :

- (1) a-i, b-ii, c-iii, d-iv
- (2) a-iv, b-iii, c-i, d-ii
- (3) a-iii, b-iv, c-i, d-ii
- (4) a-ii, b-iii, c-i, d-iv

115. How many meiotic (p) and mitotic divisions (q) are required for the development of 16 sperms from microspore mother cells in flowering plants?

- (1) $p = 8$ and $q = 16$
- (2) $p = 8$ and $q = 8$
- (3) $p = 16$ and $q = 16$
- (4) $p = 2$ and $q = 16$

116. Seeds produced by hybrid varieties will maintain hybrid traits indefinitely if seeds are produced through _____ methods.



- (1) A, C
- (2) B only
- (3) A, B
- (4) B, C

117. How many plants given below possess unisexual flowers :-
Papaya, pea, date palm, maize

- (1) Four
- (2) Three
- (3) Two
- (4) One

118. Which of the following statement is true about pollination in angiosperms ?

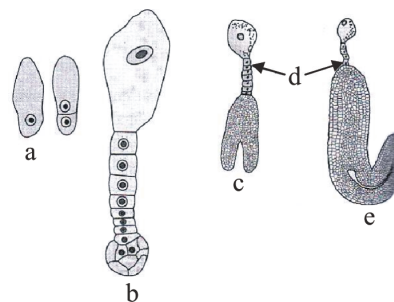
- (1) Chasmogamous flowers of *Commelina* may get cross pollinated.
- (2) Wind pollinated plants have light weight and sticky pollen grains
- (3) All the aquatic plants are pollinated by water
- (4) Waterlily is pollinated by water

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119. Recognise the figure and find out the correct matching :-



- (1) a—zygote, b—heart shaped embryo, c—globular embryo, d—radicle, e—mature embryo
- (2) a—PEN, b—globular embryo, c—heart shaped embryo, d—plumule, e—mature embryo
- (3) a—PEN, b—heart shaped embryo, c—globular embryo, d—suspensor, e—cotyledon
- (4) a—zygote, b—globular embryo, c—heart shaped embryo, d—suspensor, e—mature embryo

120. **Assertion :-** Formation of gametophyte directly from sporophyte without meiosis is called apospory.

Reason :- In apospory gametophyte always remain haploid.

- (1) Both assertion & reason are true & the reason is a **correct** explanation of the assertion.
- (2) Both assertion & reason are true but reason is **not** a correct explanation of the assertion.
- (3) Assertion is **true** but the reason is **false**
- (4) Both assertion & reason are **false**

121. Match the following (column-I with Column-II) :-

Column-I		Column-II	
a	Allele	i	Two alleles of a gene are identical
b	Genotype	ii	An individual having two different alleles of a gene
c	Homozygous	iii	Genetic constitution of an organisms
d	Heterozygous	iv	Alternative forms of a gene

- (1) a-i, b-ii, c-iii, d-iv
- (2) a-iv, b-iii, c-i, d-ii
- (3) a-iii, b-iv, c-ii, d-i
- (4) a-iii, b-ii, c-iv, d-i

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122. Which of the following trait of garden pea plant is express only in the homozygous condition.
- violet flower colour
 - yellow seed colour
 - axial flower position
 - yellow pod colour
123. A woman with albinic father marries an albinic man. The proportion of her progeny will be :-
- 2 normal : 1 albinic
 - All normal
 - All albinic
 - 1 normal : 1 albinic
124. Cross $TtRr \times TTRr$ yields $TtRR : TtRr : Ttrr : TTRr$ offspring in the ratio of
- 1 : 2 : 1 : 1
 - 2 : 1 : 1 : 2
 - 2 : 1 : 2 : 2
 - 1 : 2 : 2 : 1
125. In a cross F_1 had a phenotype that did not resemble either of the two parents and was in between the two. It represents the case of -
- Dominance
 - Co-dominance
 - Incomplete dominance
 - Complementary gene interaction
126. When I^A and I^B are present together then both express their own types of sugars, this is because of :-
- Incomplete dominance
 - Dominance
 - Co-dominance
 - Pleiotropism
127. Sickle cell anaemia is example of all except :-
- Pleiotropy
 - Multiple allelism
 - Co-dominance
 - Gene mutation
128. The height of a plant is under control of polygenic inheritance. The plant having genotype 'aabb' has 20 cm height and height of plant having genotype AABB is 60 cm, what will be height of a plant having AaBb genotype ?
- 45 cm
 - 60 cm
 - 40 cm
 - 15 cm
129. **Statement-I** : Phenotypic ratio of Monohybrid cross performed by Mendel was 3:1 in F_2 generation.
- Statement-II** : Dwarf plant recover in F_2 generation due to segregation of alleles.
- Statement-I & Statement-II both are correct.
 - Statement-I is correct & Statement-II is incorrect.
 - Statement-I is incorrect & Statement-II is correct.
 - Both Statement-I & Statement-II is incorrect.
130. Match the terms in Column-I with their description in Column-II and choose the correct option :

Column-I		Column-II	
(a)	Dominance	(i)	Many genes govern a single character
(b)	Codominance	(ii)	In a heterozygous organism only one allele expresses itself
(c)	Pleiotropy	(iii)	In a heterozygous organism both alleles express themselves fully
(d)	Polygenic inheritance	(iv)	A single gene influences many characters

- (a) - (ii), (b) - (i), (c) - (iv), (d) - (iii)
- (a) - (ii), (b) - (iii), (c) - (iv), (d) - (i)
- (a) - (iv), (b) - (i), (c) - (ii), (d) - (iii)
- (a) - (iv), (b) - (iii), (c) - (i), (d) - (ii)

131. Consider the following statements.

Statement I : In blood group the alleles I^A and I^B produce a slightly different form of sugar while allele i does not produce any sugar.

Statement II : If modified allele is produce normal enzyme then it is equivalent to unmodified allele.

- (1) Statement - I and statement - II both are incorrect
- (2) Statement - I is correct while statement - II is incorrect
- (3) Statement - I is incorrect while statement - II is correct
- (4) Statement - I and statement - II both are correct

132. Match column-I & II & select the correct options ?

	Column-I		Column-II
(i)	Complete dominance	(A)	ABO blood group
(ii)	Incomplete dominance	(B)	Coat colour in cattle
(iii)	Co-dominance	(C)	Flower colour in snapdragon
(iv)	Multiple allele	(D)	Flower colour in pea

	(i)	(ii)	(iii)	(iv)
(1)	D	C	B	A
(2)	A	B	C	D
(3)	B	A	D	C
(4)	C	A	D	B

133. Starch synthesis in the pea seed is controlled by one gene. It has two alleles B and b. If starch grain size is considered as the phenotype, then from this angle allele show :-

- (1) Multiple allele
- (2) Incomplete dominance
- (3) Co-dominance
- (4) Polygenic inheritance

134. How many genotypes and phenotypes are found in humans with respect to ABO blood group system?

- (1) 2 phenotypes and 4 genotypes
- (2) 3 phenotypes and 4 genotypes
- (3) 4 phenotypes and 6 genotypes
- (4) 4 phenotypes and 8 genotypes

135. Mendel always used pure parents for crosses, He develops pure parents by :-

- (1) Selfing of hybrids only
- (2) Hybridization process
- (3) Bagging & Tagging Technique
- (4) Continuous selfing & Selection

SECTION-B (BIOLOGY-I)

136. Which one of the following statement is correct?

- (1) Sporogenous tissue is haploid
- (2) Endothecium produces the microspores
- (3) Exine of pollen grains composed of sporopollenin.
- (4) Hard outer layer of pollen is called intine

137. All the events from pollen deposition on stigma until pollentube enters the ovule are collectively known as :-

- (1) Double fertilisation
- (2) Syngamy
- (3) Pollen Pistil interaction
- (4) Pollination

138. The scutellum observed in a wheat or maize is comparable to which part of the seed in other monocotyledons?

- (1) Plumule
- (2) Cotyledon
- (3) Endosperm
- (4) Aleurone layer

139. Which one of the following statement is not true?

- (1) Filiform apparatus present at the chalazal end of the synergids guides the entry of pollen tube
- (2) Pollen - pistil interaction is a dynamic process involving pollen recognition followed by promotion or inhibition of the pollen
- (3) Majority of insect-pollinated flowers are large, colourful, fragrant and rich in nectar
- (4) Apomixis is a form of asexual reproduction that mimics sexual reproduction

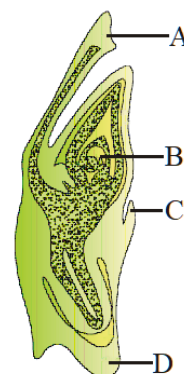
140. Haploid, diploid and triploid conditions respectively can be traced in :-

- (1) Egg, Nucellus, Endosperm
- (2) Antipodal, Egg, Endosperm
- (3) Endosperm, Nucellus, Synergids
- (4) Antipodal, Synergids, Integument

141. Smaller cell with smaller nucleus is called ____A____ and larger cell with large irregular shaped nucleus is present in mature pollen grain is known as ____B____.

- (1) A - Vegetative cell
B - Generative cell
- (2) A - Tube cell
B - Vegetative cell
- (3) A - Generative cell
B - Vegetative cell
- (4) A - Tube cell
B - Generative cell

142. Given below is the diagrammatic sketch of a transverse section of embryo. Identify the parts labelled A, B, C and D and choose the correct option about them :-



	A	B	C	D
(1)	Scutellum	Coleoptile	Radicle	Coleorrhiza
(2)	Scutellum	Shoot apex	Epiblast	Coleorrhiza
(3)	Coleoptile	Shoot apex	Cotyledon	Radicle
(4)	Coleoptile	Shoot apex	Epiblast	Plumule

143. Which of the following is not correct for prefertilization and post fertilization condition?

	Prefertilization condition		Post fertilization condition
(1)	Ovary wall	—	Fruit wall
(2)	Integuments	—	Seed coat
(3)	Endosperm	—	Perisperm
(4)	Synergid	—	Degenerate

144. **Statement I :** Removal of anthers from the flower bud during the anther dehiscence is called emasculation.

Statement II : If the female parent produces unisexual flowers, there is no need for emasculation.

- (1) Statement-I and statement-II both are true
- (2) Statement-I is true and statement-II is false
- (3) Statement-I is false and statement-II is true
- (4) Statement-I and statement-II both are false

145. How many of the following statements are true?
- (A) Embryo development precedes endosperm development in angiosperm
- (B) White kernel of coconut is free nuclear endosperm
- (C) *Vallisneria* is a dioecious plant and flowers are unisexual.
- (D) Formation of megaspore from megaspore mother cell is called megasporogenesis.
- (1) Three
- (2) Four
- (3) Two
- (4) One
146. Which of the following statement cannot be explained by dominance ?
- (1) In dissimilar pair of factors one member of the pair dominates the other
- (2) The expression of only one of the parental characters in a monohybrid cross in the F_1
- (3) Each gamete receives only one factor pair, so gametes are pure for a particular trait
- (4) The F_1 always resembled either one of the parent
147. Test cross is the cross between :
- (1) F_1 hybrid $\times$ dominant parent
- (2) F_1 hybrid $\times$ Recessive form of characters Genotype
- (3) F_1 hybrid $\times$ F_1 hybrid ($Tt \times Tt$)
- (4) F_1 hybrid $\times$ either parent
148. In *Lathyrus odoratus* which of the following cross will not produce coloured flowered offspring ?
- (1) $CCpp \times ccPP$
- (2) $CcPp \times ccpp$
- (3) $CCpp \times ccpp$
- (4) $CcPp \times ccPp$
149. **Assertion :** In sickle cell anaemia the mutant haemoglobin molecule undergoes polymerisation under low oxygen tension causing the change in shape of the RBC.
- Reason :** Heterozygous ($HbAHbS$) individuals of sickle cell anaemia appear are carrier of the disease.
- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.
150. In a dihybrid cross 3000 individuals are produced in F_2 generation. Approximately, how many will be phenotypically different from parents :-
- (1) 1250
- (2) 1000
- (3) 1500
- (4) 1125

Topic : PRINCIPLE OF INHERITANCE (GENETICS) INTRODUCTION TO CYTOPLASMIC INHERITANCE, BIOMES, BIODIVERSITY, DEMOGRAPHY AND POLLUTION, COMPLETE ANIMAL HUSBANDRY (DOA), STRUCTURAL ORGANISATION IN ANIMALS (ANIMAL TISSUE), [BACK UNIT] : LIFE CYCLE OF ANGIOSPERM

SECTION-A (BIOLOGY-II)

151. Two non-allelic gene produce new phenotype when present together but fail to do so independently are called :-

- (1) Epistasis
- (2) Polygene
- (3) Complementary gene
- (4) Pleiotropic gene

152. Exception for law of dominance are :

- (1) Linkage and Non disjunction
- (2) Incomplete dominance and Co-dominance
- (3) Incomplete dominance and Cytoplasmic inheritance
- (4) Co-dominance and linkage.

153. Which of the following statement is incorrect ?

- (1) Presence of single allele of a gene is termed as hemizygous
- (2) Two different phenotypes develops similar genotypes under different environmental condition it is known as phenocopy.
- (3) Loss of hybrid vigor due to inbreeding is known as inbreeding depression.
- (4) Recessive alleles are only expressed in homozygous condition.

154. Result of reciprocal cross changes in case :-

- (1) Genes present on sex chromosome
- (2) Both genes express themselves
- (3) Genetic expression depends on number of dominant alleles
- (4) Gene controls more than one character

155. Which statement is true about livestock and their production?

- (1) More than 70% of the world livestock population is in India and China.
- (2) India alone has more than 25% contribution to the total farm production.
- (3) China alone has more than 25% contribution to the total farm production.
- (4) China has 1st rank in milk production.

156. In MOET:

- (1) Genetic cow is administered hormones, with progesterone-like activity
- (2) The embryos are transferred at 8-32 cells stages into surrogate mothers
- (3) This is the best method for improving the production of animals which are sterile
- (4) Genetic cow releases one ovum per cycle in this technique

157. To remove the inbreeding depression, the selected animals of the breeding population should be mated with :

- (1) Related superior animals of the same breed
- (2) Unrelated superior animals of the same breed
- (3) Superior animals of the different breeds
- (4) Superior animals of the different species

158. Which one of the following is a fresh water fish?

- (1) Catla
- (2) Hilsa
- (3) Sardines
- (4) Mackerel

159. How many fishes in the list given below are marine?
Catla, Pomfret, Common carp, Hilsa, Rohu, Mackerel, Sardines

- (1) Six
- (2) Three
- (3) Four
- (4) Five

160. Identify the bird from the figure given below:-



- (1) Aseel (2) Leghorn
(3) Karaknath (4) Brahma

161. Match column I and II and choose correct option :-

Column I		Column II	
A	Adhering junction	I	Help to stop substances from leaking across tissue
B	Gap junctions	II	Perform cementing to keep neighbouring cells together
C	Tight junctions	III	Facilitate the cells to communicate each other

- (1) A-III, B-II, C-I (2) A-II, B-III, C-I
(3) A-II, B-I, C-III (4) A-I, B-III, C-II

162. Choose the correct statement for epithelial tissue:

- (1) Possesses loosely packed cells with little intercellular matrix
(2) The free surface of epithelium faces the body fluid but not the outside environment
(3) Has a basal surface and a free surface
(4) This tissue is always made of only one layer of cells

163. Which of the following statement is incorrect ?

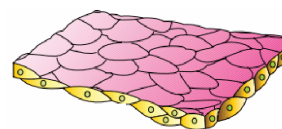
- (1) organ system split up the work in a way that exhibits division of labour.
(2) epithelium tissue has a free surface which either faces body fluids or the outside environment.
(3) some of squamous, cuboidal and columnar cells get specialised for secretion and are called glandular epithelium.
(4) main function of compound epithelium is to provide protection against chemical and mechanical stress.

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164. In multicellular animals a group of similar cells along with intercellular substances perform a specific function such an organization is called.

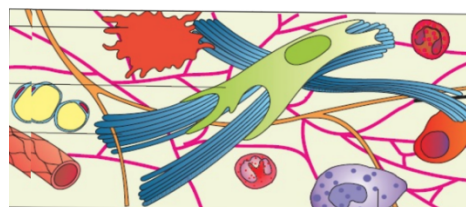
- (1) Tissue (2) Organ
(3) Body (4) Organ system

165. Select the option in which given epithelium found :



- (1) PCT (2) Wall of vessels of blood
(3) Stomach (4) DCT

166.



The above diagram is related to—

- (1) Areolar tissue
(2) Dense regular connective tissue
(3) Dense irregular connective tissue
(4) Epithelial tissue

167. Non-keratinised stratified squamous epithelium occurs in

- (1) Vagina, cervix and sweat gland ducts
(2) Vagina, skin, nails
(3) Skin and nails
(4) Vagina, cervix, buccal cavity

168. The most widely distributed tissue in the body of complex animals :-

- (1) Epithelial tissue
(2) Connective tissue
(3) Muscular tissue
(4) Nervous tissue

169. The inter cellular material of ___'A'___ is solid and pliable and resist compression. Identify 'A'.

- (1) Cartilage (2) Bone
(3) Blood (4) Muscle

170. Most of the cartilage in vertebrate embryos are replaced by ___'X'___ in adults identify X.

- (1) Dense connective tissue
(2) Loose connective tissue
(3) Bones
(4) Cartilage

171. **Statement-A** : Bones have hard and non-pliable ground substance.

Statement-B : Bones are rich in calcium salt and collagen fibre which give bone its strength.

- (1) Both statements are correct
(2) Both statements are incorrect
(3) Statement A is incorrect
(4) Statement B is incorrect

172. **Statement-A** : The bone cells (chondrocytes) are present in the space is called lacuna.

Statement-B : Cells of the cartilage (osteocytes) are enclosed in small cavities within the matrix secreted by them.

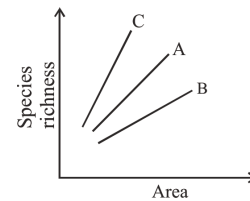
- (1) Both statements are correct
(2) Both statements are incorrect
(3) Statement A is only correct
(4) Statement B is only correct

173. India is the one of the 12 mega diversity countries of the world, it shares what percentage of global species diversity?

- (1) 3.4%
(2) 12%
(3) 6%
(4) 8.1%

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174. According Alexander von Humbolt, in the graph of species Area relative of area A, B and C which of the following area has steeper slope and minimum species richness respectively :-



- (1) A and B (2) C and A
(3) C and B (4) C and C

175. Resistant and resilient to occassional disturbances (natural or man-made) and resistant to invasions by alien species. It is a character of :-

- (1) Unstable community with more species and high biomass and productivity
(2) Unstable community with less biomass and less productivity
(3) Stable community with more species, less biomass and less productivity
(4) Stable community with more species, high biomass and high productivity

176. Which among the following are exotic/alien species introduced in India :-

- (a) Parthenium hysterophorous
(b) Lantana camara
(c) *Eichhornia crassipes*
(d) *Clarias gariepinus*

- (1) b,c,d (2) a,b,c
(3) a,c (4) a,b,c,d

177. Which one of the following is not included under in situ conservation ?

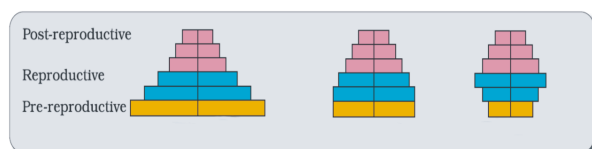
- (1) National park
(2) Wild life sanctuary
(3) Cryopreservation
(4) Biosphere reserve

178. Assertion : In an area, if there are 200 *Parthenium* plants and one huge banyan tree with large canopy, biomass is more meaningful measurement of N (Population Density).

Reason : In this case, number is meaning less which underestimates the enormous role of banyan tree in that community.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

179. Identify the growth status of the population by the following the age pyramids :-



	A	B	C
(1)	Expanding	Declining	Stable
(2)	Expanding	Stable	Declining
(3)	Stable	Expanding	Declining
(4)	Declining	Stable	Expanding

180. Which one of the following pair of organism produce a large number of small sized offspring ?

- (1) Pacific Salmon Fish, Bamboo
- (2) Oysters, Pelagic Fish
- (3) Oysters, Pacific Salmon Fish
- (4) Bamboo, Pelagic Fish

181. The specific intensity of Noise, which may damage ear drums and causes permanently impairing hearing ability is ?

- (1) Equal and less than 150 dB
- (2) Equal and more than 150 dB
- (3) More than 200 dB
- (4) More than 100 dB

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182. Match the items given in Column I with those in Column II and select the **correct** option given below:

	Column I		Column II
a.	Eutrophication	i.	UV-B radiation
b.	Sanitary landfill	ii.	Deforestation
c.	Snow blindness	iii.	Nutrient enrichment
d.	Jhum cultivation	iv.	Waste disposal

	a	b	c	d
(1)	ii	i	iii	iv
(2)	i	iii	iv	ii
(3)	iii	iv	i	ii
(4)	i	ii	iv	iii

183. "Green muffler" technique is related to control of which pollution :

- (1) Soil (2) Water (3) Noise (4) Air

184. In logistic growth curve, asymptote is obtained when :

- (1) $K = N$
- (2) $K > N$
- (3) $K < N$
- (4) value of 'r' approaches zero

185. Main driving force of species extinction is :

- (1) Habitat loss and fragmentation
- (2) Co-extinction
- (3) Alien species invasion
- (4) Over-exploitation

SECTION-B (BIOLOGY-II)

186. A child with blood group genotype $I_A I_B$ is form of a woman with genotype $I_B I_B$. The father could not be a man of genotype :-

- (1) $I^A I^A$
- (2) $I^A I^B$
- (3) $I^B I^B$
- (4) $I^A i$

187. Correct sequence of Mendel's experiment is :-

- (1) Emasculation → Bagging → Tagging → Dusting
- (2) Selfing of F_1 → Emasculation → Bagging → Tagging
- (3) Dusting → Tagging → Bagging → Emasculation
- (4) Selection of pure parent → Selfing of F_1 → Cross pollination of F_2 .

188. Out-crossing is carried out between the animals of:

- (1) Same breed
- (2) Different breed
- (3) Different species
- (4) Different genera

189. Which mating should be done to combine the quality of two different breeds in farm animals:

- (1) Inbreeding
- (2) Out-crossing
- (3) Cross-breeding
- (4) Interspecific hybridization

190. Inbreeding can't be used to :

- (1) Obtain purelines
- (2) Increase homozygosity
- (3) Combine the qualities of two different breeds
- (4) Increase productivity

191. Select the incorrect match :

- (1) Squamous epithelium – Irregular boundaries
- (2) Cuboidal epithelium – Duct of glands
- (3) Columnar epithelium – Elongated central nucleus
- (4) Ciliated epithelium – Bronchioles and fallopian tube

192. **Assertion** : Gap junctions facilitate the cells to communicate with each other by connecting the cytoplasm of adjoining cell's.

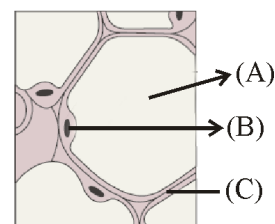
Reason : Gap junction helps rapid transfer of ions, small molecules and sometimes big molecules.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

193. The supportive skeletal structures in the human external ears and in the nose tip are examples of :-

- (1) bone
- (2) cartilage
- (3) ligament
- (4) areolar tissue

194. Identify A, B and C in the given diagram of adipose tissue.



- (1) A-Cytoplasm, B-Nucleus, C-cellwall
- (2) A-Fat storage area, B-mast cell, C-plasma membrane
- (3) A-cell fluid, B-collagen fibres, C-plasmalemma
- (4) A-Fat storage area, B-nucleus, C-plasma membrane

195. Assertion :- There is no clearcut estimation about biodiversity of prokaryotes.

Reason :- Conventional taxonomic methods are not suitable for identification of microbes and many species are not culturable.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

196. Which of the following are correct statements ?

- (A) Number of Fungi is more than total vertebrates
- (B) Constant environment promote niche specialization which leads to greater species diversity
- (C) Species inventories are more complete in tropics than temperate
- (D) Rich biodiversity is not only essential for ecosystem health but imperative for very survival of human race on this planet.

- (1) A, B, C, D
- (2) A, B, D
- (3) A, C, D
- (4) A, B, C

197. Match these columns :

	A		B
(a)	Dodo	(i)	Africa
(b)	Quagga	(ii)	Russia
(c)	Stellar sea cow	(iii)	Bali, Java, Caspian
(d)	Thylacine	(iv)	Australia
(e)	Tiger sub-species	(v)	Mauritius

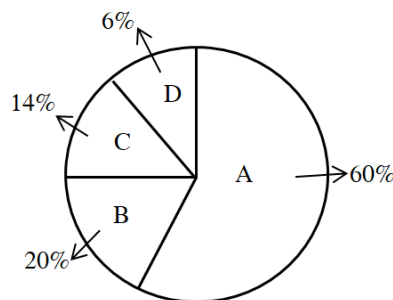
	(a)	(b)	(c)	(d)	(e)
(1)	i	ii	iii	iv	v
(2)	v	iv	ii	i	iii
(3)	v	i	ii	iv	iii
(4)	ii	v	iii	i	iv

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PHASE - MEB

18-06-2023

198.



Contribution of green house gases given. Identify correct match.

- (1) A - It forms carboxyhemoglobin
- (2) B - Laughing gas
- (3) C - Ozone depleting substance
- (4) D - Marshy gas

199. Read the following statements :

- (A) Effluents from industries and homes can radically accelerate the aging of lake, is called Accelerated Eutrophication.
- (B) BOD is a measure of organic matter present in the water.
- (C) Bacterial flocs when sediment, is called Activated Sludge.
- (D) Non-biodegradable pollutants accumulate in tissues in increasing concentration along the food chain is called Biomagnification.

Choose correct ones :

- (1) A, B, C
- (2) A, B, D
- (3) B, C, D
- (4) A, B, C, D

200. Read following statements and choose incorrect ?

- (1) Scrubber can remove gaseous pollutant like SO₂
- (2) Electrostatic precipitator remove 99% of particulate matter from thermal power plant exhaust.
- (3) Catalytic converters are beneficial for removing pollutants from industrial emissions.
- (4) 50 ppm sulphur amount in petrol and diesel in BSIV.

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 ALLEN[®] CAREER INSTITUTE KOTA (RAJASTHAN) QUESTIONS BREAK-UP CHART FOR MINOR TEST FROM CURRENT & PREVIOUS UNITS (Session : 2023 - 2024)			
ENTHUSE ADV. COURSE : MEB			
MINOR TEST NO. & DATE	CURRENT UNIT	PREVIOUS UNIT (COMPLETE TOPIC)	
06 16-07-2023	SYLLABUS COVERED FROM 19-06-2023 TO 15-07-2023	P	CAPACITOR & GRAVITATION
		C	CHEMICAL BONDING
		B	HUMAN HEALTH & DISEASE

TG: @Chalnaayaaar

PHASE - MEB

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18-06-2023

